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Towards eXplicitly eXplainable Artificial Intelligence

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ABSTRACT

Artificial Intelligence (AI) plays a leading role in Industry 4.0 and future Industry 5.0. Concerns about the opacity of today's neural network AI solutions have led to the Explainable AI (XAI) project, which attempts to open the black box of neural networks. While XAI can help to partially interpret and explain the workings of neural networks, it has not changed their original subsymbolic nature and the opaque statistical nature of their workings. Significant uncertainties remain about the safety, reliability, and accountability of modern neural network AI solutions. Here we present a novel AI method that has a fully transparent white-box nature - eXplicitly eXplainable Artificial Intelligence (XXAI). XXAI is implemented on deterministic cellular automata whose rules are based on first principles of the problem domain. XXAI overcomes the limitations for a broader application of symbolic AI. The practical value of XXAI lies in its ability to make autonomous, fully transparent decisions due to its multi-component, multi-level, networked, hyper-logical nature. Looking ahead, XXAI has the potential to become a leading strategic partner in the field of neuro-symbolic hybrid AI systems. XXAI is able to systematically validate neural network solutions, ensuring that the required standards of reliability, security and ethics are met throughout the AI lifecycle, from training to deployment. By creating a clear cognitive framework, XXAI will enable the development of advanced autonomous solutions to achieve the human-centric values of the future Industry 5.0. A comprehensive program for the further development of the proposed approach is presented.

1. Introduction

1.1. AI and industry 4.0 - 6.0

Artificial Intelligence (AI) plays a leading role in Industry 4.0, the upcoming Industry 5.0 [1], and the futuristic Industry 6.0 [2]. Industry 5.0 will bring a deep human dimension to industrial automation by encouraging cooperation between humans and robots, connecting human cognitive capabilities with smart manufacturing. It will restore the importance of the human aspect and bring back the human dimension to manufacturing by encouraging cooperation between humans and robots, in addition to the technological advances developed in Industry 4.0 [1]. The futuristic Industry 6.0 will aim to create fully integrated, intelligent manufacturing systems that can operate autonomously with minimal human intervention while achieving shared semantic hyper-connectivity [2].

1.2. The AI opacity problem: an unprecedented challenge to human intelligence ahead of industry 5.0

Today, the greatest achievements in AI are associated with eXplainable Artificial Intelligence (XAI) and especially its generative version, GPT-4. In addition to outstanding achievements, modern artificial intelligence has a number of frightening drawbacks. The main drawback is the opacity of neural network AI solutions to human understanding. The basis of neural network AI solutions cannot be expressed in the logical form of cause and effect. Only an unreliable indirect explanation of how modern neural network AI works within the framework of XAI is possible. The opaque nature of neural network AI as a black box remains unchanged. As a result - we cannot be sure of the safety of autonomous solutions of modern AI. Considering the very fast growth of the used computing power, the opacity of AI solutions causes panic fears up to demands to stop the further improvement of generative AI completely [3]. Attempts to limit the capabilities of neural network AI by introducing human control seem naive, as they contradict the very

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1566-2535/© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

idea of automation. Moreover, human control is fundamentally limited by the extreme superiority of AI in speed and volume of information processing. It is obvious that the control over the decisions of opaque artificial intelligence should be taken over by a fully transparent symbolic AI of broad purpose, the creation of which has been greatly delayed. All this poses a cognitive challenge to the human mind not seen since the Renaissance [4].

1.3. Deafening wall of incomprehension due to intellectual debt in AI

There is a deaf wall of misunderstanding between humans and AI - AI does not understand human knowledge in its symbolic-rational form, perceiving the initial data only as a statistical ensemble of images, and humans are not able to understand the logical cause-and-effect bases of AI decisions because they have an opaque, non-logical nature. The reason for this alarming situation is that modern machine learning systems are built with an unacceptable intellectual debt [5–7]. This unacceptable intellectual debt has its roots in the peculiarities of modern science, which, with the reign of positivism in the 19th century, effectively abandoned the classical scientific method and limited itself to the accumulation and mathematical processing of experimental data. People have reflexively relaxed and stopped thinking about many questions, entrusting their solutions to the results of experiments, to opaque mathematical models [8], and to the services of modern opaque AI. It is practically forgotten that the human intellect is capable on a holistic level of performing idealizing information fusion of experimental data and creating general abstract theories underlying specifically human adaptation to the environment. It is classical rational theories that form the basis of today's human technological power. To remedy the situation, humanity must invest in the redemption of the collective intellectual debt.

1.4. Cognitive foundation for transparent AI

Human intelligence is the result of billions of years of living systems adapting to environmental conditions. The smartest and most creative evolved to the top of the evolutionary ladder. This evolutionary basis is supplemented by a personal inclination toward science and a broad university education. Specifically, human adaptation to the world occurs through the creation of theory as a way of finding solutions to a problem situation. Faced with a problem situation, a person begins to analyze it - to break it down into elementary objects, to find out the rules (axioms) of relations of these elementary objects, to check and recheck the correctness of these basic building blocks, and then to reconstruct the situation in a more perfect form as a way out of the problem. Rational theories are formalized as a coherent set of basic objects and axioms for a given problematic domain. These objects and axioms are the first principles of science. Having a theory warns us in advance of problems when the theory stops working. That means we need a new theory. The human mind is the perfect tool for abstracting the data of experience and constructing new knowledge from the resulting abstract objects. Rational knowledge is the foundation of understanding. Understanding implies the possibility of comprehensive rational reproduction of the studied object in a person's head, in a computer, and further in a physical construction. Objects and axioms of scientific theories are simple and obvious for understanding by different people due to similarity of practical experience and common intellectual intuition. Therefore, axiomatic theories serve as the basis for the most effective transfer of knowledge from human to human and from human to artificial intelligence. Axiomatic theories are transparent and mechanistic, so they can be used to design optimal behavior and efficient engineering devices. The successes of scientific rationality have brought so many benefits that, out of habit, we have begun to call any useful solution rational. However, the shift to understanding rationality as practical utility should not make us forget that scientific rationality is based on theoretical supersensible (abstract) concepts.

Classical scientific rationalism's quest for the most general knowledge has been actively thwarted by the various forms of positivism that dominate today. Positivism in all its forms seeks to free science from the search for understanding the essence of systems, which is incomprehensible to its adherents. In fact, understanding the essence of a system is simply creating a transparent model of it, which defines the basic elementary objects of the system and the rules of their interaction (axioms) within the system as a whole. Positivism insists that the only source of scientific knowledge is empirical research and that all scientific hypotheses arise by chance. Positivists believe that the axioms of scientific theories are arbitrarily formulated and deny the value of scientific reflection, calling it speculation. The dominance of positivism in science has led to work on a general picture of the world being seen as unscientific. Science from a highly spiritual sphere began to turn into a kind of experimental craft activity. It was forgotten that the foundations of all sciences, including the atomic theory and axiomatic geometry of Euclid, were created by the ancient Greeks on the basis of mental experiments. We believe that to create a broad cognitive foundation for transparent AI, it is necessary to revitalize the classical rational scientific method and broad classical education.

In this paper, we analyze the problematic situation in modern AI and propose our solution to create fully transparent AI as a concrete way to eliminate the accumulated cognitive debt in anticipation of Industry 5.0. We have named this algorithmically transparent artificial intelligence "eXplicitly eXplainable AI" (XXAI). XXAI makes decisions as automatic causal reasoning and has the potential to become a senior partner to insufficiently transparent eXplainable AI (XAI) in hybrid solutions.

1.5. Concerns about reliability and security of contemporary AI

Machine learning systems based on artificial neural networks have achieved a number of breakthroughs by combining huge machine learning data sets with the enormous processing power of modern computers. However, it is well known that machine learning AI based on neural networks has a "black box" nature and its decisions cannot be explicitly explained. Concerns about the trustworthiness and security of their solutions have led to the AI black box problem. The mathematical models of complex systems that underlie today's most successful AI systems are based on methods from continuous mathematics and probability theory. These models are of the "black box" type. At the same time, our world clearly consists of discrete systems. The computers we use are also discrete in nature. Much effort has gone into making discrete computers suitable for continuous mathematical modeling. Opaque mathematical models of the black box type are not able to take into account the local conditions of the behavior of the subsystems of the objects under study and are based rather on fitting the model parameters to the observed phenomenology or known facts. The conclusions of opaque models about the internal mechanisms of complex systems have to be made as indirect subjective interpretations, which gives rise to paradoxes and complicates the understanding of the phenomena under study. All this is true for the most successful neural network AI of the GPT type today, based on the Transformer model. These new approaches to deep learning have overshadowed all earlier approaches to AI [9]. However, Transformer models, including GPT versions, are opaque and do not yet create human-interpretable semantic representations that underlie generated sentences and documents. Transformer models do not use pure logic operations such as AND, OR, XOR and they do not rely on production rules such as "if-then" rules. The algorithms of the Transformer model allow it to make decisions about which word or phrase should follow the previous words or phrases in order to generate the text with the highest likelihood of believability. At the same time, AI based on the Transformer architecture has a higher degree of explainability of the decisions made compared to classical neural networks. This is because Transformer models use an attention mechanism that allows you to understand what input was used to generate each part of the output. Thus, Transformer models can provide some explanations and

interpretations of their predictions. However, as with other deep learning models, complete transparency is still an elusive goal, and some aspects of the model may not be explicitly explainable. Decisions of machine learning systems are made on an irrational, sub-symbolic basis. We cannot know the logic of the decisions they make, and we learn these decisions only after the fact from the behavior of the AI-controlled system. This lack of transparency in decision making can lead to errors, especially in unfamiliar, unexpected, and unusual situations. Douglas Lenat, who has led the development of the largest knowledge base of common sense "CYC" since 1984, compares modern AI with a patient whose left hemisphere of the brain, which is responsible for logical thinking, is completely damaged [10]. These circumstances are unacceptable for a number of sensitive industries – medicine [11,12], law, transport, insurance, security systems, including military ones. The opacity of modern AI solutions is compounded by the added complexity of these sensitive industries. Thus, in the paper [13] highlights the difficulties AI faces in making ethical decisions when individual interests conflict with the public good, as exemplified by autonomous vehicle scenarios. The authors also explore the legal challenges, emphasizing the need for a regulatory framework that addresses the complex issues of liability and safety in AI applications while not hindering innovation. The paper [14] discusses how the biases inherent in AI training data contribute to existing social inequalities, particularly in the labor market. This negatively impacts opportunities and social mobility in areas such as justice, health, education, and human resource management. The study highlights the importance of using diverse and representative datasets to mitigate bias and ensure equity in AI-based decision-making. The paper [15] investigates how language influences human decision making in game-theoretic contexts and proposes a shift from traditional outcome-based utility functions to functions that incorporate language preferences. The authors argue that language design significantly influences decision making. This study lays the groundwork for integrating AI-generated language into decision-making models, emphasizing the importance of sentiment in shaping human preferences and interactions with AI systems. We believe that the combination of operational and semantic transparency capabilities in our proposed XXAI will help address the issues raised in [13–15].

In 2016, DARPA, the U.S. Department of Defense's Defense Advanced Research Projects Agency, initiated the Explainable Artificial Intelligence (XAI) project [16]. The purpose of the XAI Project is to "open the black box" of machine learning in order to explain the reasons for the decisions made by AI, to increase the security and reliability of these decisions, and to increase the trust of the end user in them. The strategic goal of the project is to create autonomous intelligent systems that perceive, learn, make decisions and act independently, symbiotically combining machine learning with other artificial intelligence technologies. In subsequent years, a lot of research on "XAI" appeared, but so far, they are not devoted to opening the black box of neural network AI, but to assurances that black box systems can be trusted [11, 17–21]. The authors [22] propose a new XAI framework, called "Symbolic XAI", to explain machine learning models in a symbolic manner by creating human-understandable logical relationships between the input features and the outputs of the model. The "Symbolic XAI" framework is an important contribution to the field of eXplainable AI and has promising potential. However, it fully preserves the core problem - the operation of the neural network itself remains unchanged, retaining its subsymbolic nature and the opaque statistical nature of its operation. These are attempts to legitimize black box AI, not to make AI truly transparent. The black box of neural network AI remains closed. We agree with the opinion [12] – "Some people hope that creating methods for explaining these black box models will alleviate some of the problems, but trying to explain black box models, rather than creating models that are interpretable in the first place, is likely to perpetuate bad practice and can potentially cause great harm to society". Dissatisfaction with the results of attempts to crack the AI black box in the XAI project led to the need to go beyond the scope of that project and move directly to ensuring transparency, efficiency, and reliability - [23].

Let us summarize the main problem with today's XAI. While XAI can help partially explain how neural networks work, it has not changed their subsymbolic nature and the opaque statistical nature of their operation. There remains considerable uncertainty about the safety, reliability, and accountability of existing neural network solutions.

1.6. Four fundamental barriers to the widespread use of "white box" AI

Attempts to crack the black box of contemporary AI and find a logical explanation for its decision-making processes are reminiscent of attempts to capture thoughts in the human brain with a scalpel and electrodes. We believe that instead of trying to crack the black box of AI, we need to create its fully transparent partner, which is inherently a white box. Really explicable and transparent artificial intelligence with available decision-making algorithms should use white-box mathematical models [8,24]. Unlike black-box AI, white-box AI decision algorithms are human-understandable and can be based on ethical, usefulness, and safety principles. In the future, fully transparent AI based on logical cause-and-effect reasoning can effectively complement non-transparent machine learning to form a hybrid neuro-symbolic AI [25–35].

Today, the learning process of deep neural network transformers using the Reinforcement Learning from Human Feedback method is done through human mediation [36]. or using the Reinforcement Learning with AI Feedback (RLAIF) method mediated by another transformer neural network [37]. Human control of the neural network training process is expensive and time-consuming. In addition, human control over AI decisions makes the very idea of automation largely meaningless, and human capabilities in terms of speed of decision-making and volume of information processed are significantly inferior to the capabilities of modern computers. We believe that in future of hybrid artificial intelligence, symbolic AI will take control of the learning and decisions of sub-symbolic AI.

Norbert Wiener, in his Cybernetics, used the terms "black box" and "white box" to characterize types of mathematical models of complex systems [38]. While mathematical "black box" models are now well known and successfully applied, the implementation of "white box" models, especially in artificial intelligence, still raises questions. We treat phenomenological models of complex systems as black boxes and mechanistic models of complex systems as white boxes [39,40]. Fully mechanistic models of complex systems are completely discrete and deterministic. They allow us to directly gain a mechanistic insight into a system under study [24,41,42].

Symbolic AI is a completely transparent white box. It allows logical verification of the security and reliability of its solutions. Symbolic AI works logically with symbols. Each symbol in symbolic AI represents a specific concept. This provides a causal understanding of its decisions. Understanding is our ability to logically reproduce the decision process in our imagination, in text, or in a computer. Such reproduction begins with a specific element base of the model and takes into account all local connections of all elements within the model, allowing the entire model to be reproduced as a whole. Decision clarity is characterized by its transparency, which allows symbolic AI to be viewed as a white box. At the moment, the most successful implementations of artificial intelligence are its subsymbolic forms based on neural networks. Subsymbolic AI is a black box because humans cannot logically trace the cause-and-effect mechanisms of their decisions. This is the black box problem of the modern AI [43]. To be as good at solving a variety of complex problems as humans are, machines must learn to build logical cause-and-effect models of the environment and navigate different contexts. Building logical cause-and-effect models is a specialization of symbolic AI.

In the early days of AI, logical and symbolic reasoning methods predominated [44,45]. AI of a symbolic-logical nature is known as Good Old-Fashioned Artificial Intelligence (GOFAI) [46,47]. The birth of

artificial intelligence and expert systems in the 1960s and 1970s led to significant progress in the field of symbolic AI. From the 1970s to the 1990s, these symbolic AI techniques faced a number of challenges that were never overcome.

Let's list four fundamental barriers that were identified half a century ago and still prevent the widespread adoption of symbolic AI:

1. *Operational opacity of the mathematical apparatus*. This problem arose due to the lack of transparency of the most mathematical methods, which are fundamentally unable to control the local interactions of subsystems of the modeled system [24,39,40,42].
2. *Semantic (ontological) opacity* (implicitness, non-strictness and polysemousness) *of natural language terms* that lead to *general abstract ontology* errors in automatic inference [48–54].
3. *Lack of a general abstract ontology of our world.* This does not allow realizing a common scientific understanding, agreeing on the meanings of concepts, and formulating universal ethical principles that are common to humans and AI. This general top-level ontological knowledge should be formulated in explicit, easily applicable definitions [52,55,56].
4. *Irresistible combinatorial explosion.* Often an irresistible combinatorial explosion that occurs when sorting through logically possible options in search of the best solution [57]. Such an enumeration may require more time than the time of the heat death of our Universe [10].

Failure to solve any of these problems will negate the widespread use of symbolic AI. The safe, secure and trustworthy development and use of artificial intelligence is receiving great attention [58,59]. The quest to ensure the reliability and security of the most effective AI faces the challenge of ensuring the algorithmic transparency of the decisions made. Only symbolic AI, through its full algorithmic transparency, can implement automatic protection of AI decisions to keep pace with existing and emerging threats. Let us characterize these obstacles and ways to overcome them in more detail.

2. The way to an eXplicitly eXplainable artificial intelligence

Main hypothesis

Explicitly Explainable AI (XXAI) is possible as a white box solution to the AI black box problem that overcomes all four barriers to the widespread use of symbolic AI listed here on the basis of deterministic logical cellular automata whose rules are based on the axioms (the first principles) of the general physical theory of the relevant domain:

1. *Operational (syntactic) transparency* of the mathematical apparatus is achieved by cellular-automatic cause-and-effect fusion of all local cause-and-effect inferences into a global causal conclusion. These global conclusions are accepted at each iteration of the cellular automaton.
2. *Semantic (ontological) transparency* is achieved through the use of explicit concepts both at the level of local decisions about the behavior of individuals of the element base, and at higher levels of organization, including the behavior of the system as a whole through the use of the axioms of the general theory of the relevant domain.
3. *The general abstract ontology of our world* is based on a general axiomatic theory of life [60,61].
4. *The insurmountable combinatorial explosion in the search for options is* eliminated by direct cause-and-effect cellular automata aggregation of local solutions into a single global solution. The aggregation is based on cellular automata neighborhood and cause-and-effect rules of state transitions. Scenarios for the implementation of these cause-and-effect rules are developed on the basis of the axioms of the general physical theory of the relevant domain.

XXAI has the potential to become a leading strategic partner in the field of neuro-symbolic hybrid AI systems. XXAI is able to systematically validate neural network solutions, ensuring that the required standards of reliability, security and ethics are met throughout the AI lifecycle, from training to deployment.

2.2. Overcoming the barriers to the widespread use of the "white box" AI

To create a general-purpose symbolic AI, it is necessary to implement a transparent mathematical algorithm, i.e. the ability of the AI to make decisions algorithmically, hierarchically combining individual logical steps into a common output. The ability to implement a complex algorithm as a whole, starting from its elementary components, is based on the knowledge of the mechanism of operation of the system. This knowledge is the foundation of our understanding. It is this understanding of AI decision algorithms that underlies their explicability. Next, we take a closer look at ways to overcome the barriers to widespread adoption of symbolic AI, which can work with knowledge and make decisions we can fully understand.

Overcoming operational opacity of basic mathematical models

The paper [62] points out the need to create global frameworks that integrate local cause-and-effect interactions to solve the black box problem in AI. Cellular automata allow you to create a transparent model of the system, providing integration of the behavior of the system elements through their local interactions. *Logical deterministic cellular automata are an uncontested basic tool for ensuring the operational transparency of mathematical models for AI.* The transparency of such models is ensured by the ability to control the states and relationships of basic elements within the modeled system through a local cellular automaton neighborhood. We can talk about a special cellular-automaton method of symbolic thinking, which combines the following two points: (1) space modeling by a grid of discrete states of the element base of the system; (2) modeling time by successive states of system elements according to the rules of local dependencies specified by the general theory of the corresponding subject area.

Fig. 1 shows our classification of mathematical models according to their degree of transparency, depending on the type of mathematical methods used [39,41].

It follows from the hypothesis presented by us that a completely transparent model of a system may be created by logical deterministic cellular automata, the rules of which are based on the general physical theory of the subject area of the system being modeled.

2.2.2. Overcoming semantic opacity (implicitness, non-strictness and polysemy) of natural language terms

Natural language concepts present inherent limitations, including their implicit nature, insufficient semantic coordination, and potential errors and contradictions in conventional ontologies, which impede the development of reliable automated reasoning systems.

The semantic explicitness of terms can be achieved only within the framework of the general theory of the relevant subject area. Within the framework of such a theory, the terms not only have a strictly defined meaning, but their meanings are fully consistent with each other at all levels of the organization of the model.

To ensure correct automatic inference, it is necessary to carry out the procedure of preliminary theoretical purification (idealization) of the used concepts of natural language, applying the principles of classical scientific rationality. The mysterious power of mathematics [63] lies precisely in the fact that they are based on a strict axiomatic definition of basic concepts, rules for operating these basic concepts, and on careful coordination of these concepts into a single integral theory. Such a theory is the most perfect knowledge base for recreating the objects of the relevant subject area in the minds of students, in computer models and in technical objects, including at a new, more advanced level.

To understand means to reproduce an object in the imagination or in

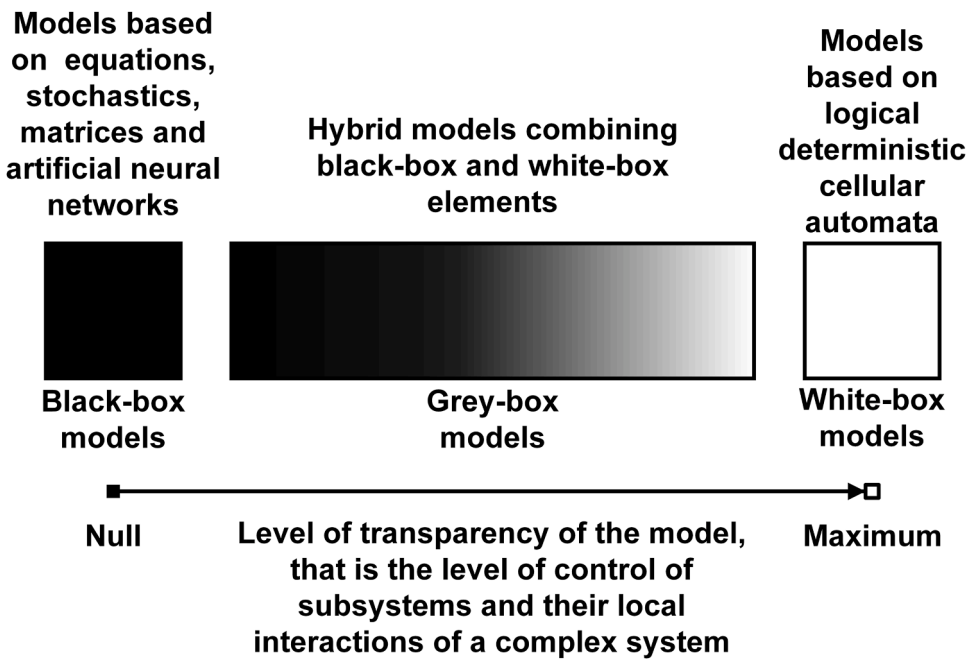

Fig. 1. Classification of mathematical models of complex systems according to the transparency of the mathematical methods used: "black box", "gray box" and "white box". While "black box" models lack mechanistic understanding because they cannot account for local causal interactions of subsystems, "white box" models provide the full mechanistic understanding needed to ensure safety, reliability, and accountability of AI in highly adaptive and autonomous industrial environments.

a computer, starting from its elemental base. Scientific rational knowledge provides such an opportunity. Here a cultural problem arises - rational knowledge has a supersensory abstract nature, and today the acquisition of experimental data and their statistical processing are at the forefront of science. Abstract results are often considered speculative and are more likely to be disapproved of by reviewers. John Maddox, editor-in-chief of Nature, noted that publishing Watson and Crick's famous paper on the structure of DNA would not be possible today because reviewers would surely find it speculative [64]. This article was published without peer review as self-evident [65].

2.2.3. Creation of a common abstract ontology of our world

We are convinced that the creation of an ontology of transparent AI is possible only on the basis of a general abstract theory of the relevant subject area. For successful work in this direction, the culture of creating such general abstract theories must be comprehensively developed. The experience of mathematics and programming in formal languages has shown that logical operations with symbols corresponding to completely explicit concepts give semantically correct results. To turn partially implicit concepts into explicit ones, it is necessary to build a general axiomatic theory of the subject area of the modeled object. Such a detailed specification of conceptualization is the construction of an ontology of a given subject area [66]. Each specific axiomatic theory makes it possible to build an adequate logical model in its subject area. For general artificial intelligence, it will be necessary to create a general ontological theory that is equally applicable to any subject area. In our opinion this problem coincides with the general formulation of Hilbert's sixth problem on the existence of mathematical foundations of the physical world. At present, there is no formal theory in which to express the analysis of the means and ends used both in ordinary life and in physics. The creation of such a formal theory of the most general nature is one of the key promising tasks for ensuring the reliability of symbolic AI [67]. A common abstract ontology provides a unified conceptual framework for different domains. For our research, the general semantic basis was the abstract theory of living systems [60,61]. Based on this theory, we formulated a general axiomatic theory of the ecosystem for the study of interspecific competition, which is presented in our previous works [40,41,68] and in most detail in our article [24].

2.2.4. Solving the combinatorial explosion problem

The fourth problem faced by the developers of symbolic intelligence in the 70 s was the often-insurmountable combinatorial explosion that occurs when a complete enumeration of possible options in search of the best solution [57]. To overcome the problems of complete enumeration of the space of possible solutions, we use the direct derivation of the optimal solution, starting from the local rules for the optimal behavior of micro-objects of the simulated system. Logical local rules for the behavior of micro-objects are given by the cellular automaton neighborhood and glue the behavior of micro-objects into the behavior of an integral system. The evolution time of such a model as a whole is directly proportional to the size of the system and the length of the rules of behavior used. Local cellular automaton rules should be based on the first principles of the general theory of the subject area. Then the operational cellular automaton model building from the bottom up is complemented by the simultaneous semantic model building from the top down. As a result of combining the possibilities of the general abstract theory of the subject area of the modeled object with the possibilities of productive cellular-automatic inference, automatic deductive inference is carried out. A detailed description of the method is presented in our work [24].

3. Discussion of the problems of symbolic AI and ways to overcome them

3.1. Features and capabilities of cellular automata

"Cellular automata are mathematical idealizations of physical systems in which space and time are discrete, and physical quantities take on a finite set of discrete values" [69]. The system is modeled by a cellular automaton as an iteratively changing structured set of homogeneous elements. Each element of this set can be in at least two states. The states of all elements of this set change step by step and

simultaneously according to certain logical rules. The rules determine what state each given element will go to at the next iteration site and depend on what state the element itself is in at the moment and what states its neighbors are in, given by the local cellular automaton neighborhood. Homogeneous elements, coordinated with each other locally throughout the space of a structured set, are always and uniquely glued together into an integral global system. The global behavior of the system is generated from the local rules for changing the states of its micro-objects.

Formally, a cellular automaton can be defined as a set of five objects:

1. Regular homogeneous lattice of sites (each site models a micro-object);
2. A finite set of possible site states;
3. Initial pattern of states of lattice sites.
4. A neighborhood consisting of a site and certain neighboring sites. This set of sites affects each site transition to the next state;
5. The function of transition to the next state - the rules of iterative change of states of each site without exception, which are applied to all sites of the lattice simultaneously. These rules take into account the state of the neighborhood of each site. All sites have the same update rules. The site state iterations model the timeline. The logical causal structure that governs the transition between site states is based on if-then rules.

Each site of a cellular automaton is a finite automaton, and the cellular automaton as a whole is a polyautomaton that co-organizes the behavior of the finite automata located in the sites of its lattice. The cellular automaton implements automatic inference as hyperlogic, since the cellular automaton logic is executed simultaneously for all sites at each iteration. As a result, the cellular automaton implements knowledge-based inference. The behavior of the micro-objects of the simulated system is logically glued together into a single integral macro-object with the help of local interactions mediated by the cellular automaton neighborhood. It is this kind of modeling that logically reproduces the integration of subsystems into an integral system that allows us to speak about the transparency of models.

3.2. Our Universe is a cellular automaton

The deterministic dynamics of cellular automata appears to be a candidate for the fundamental laws of physics, including quantum theory [70]. The effectiveness of cellular automaton modeling can be related to the fact that the fundamental laws of nature are of a cellular automaton nature, and the Universe itself is a cellular automaton. Such a hypothesis was put forward earlier by Konrad Zuse in his book "Computing Space" [71]. Zuse calls for an "automatic way of thinking" in the field of physics, which implies that a physical model is considered in terms of a sequence of states following each other according to predetermined rules. Following Zuse, Edward Fredkin believed that the Universe is a computer program that develops on a grid in accordance with a simple rule [72]. If we could only define the right grid type and the right rule, Fredkin says, we could model the universe and all of physics. Fredkin insists that the rule governing the universe's cellular automaton should be simple—just a few lines of clean and elegant code. Continuing the ideas of K. Zuse and Fredkin, Stephen Wolfram made attempts to find a digital cellular automaton code of the Universe. Eighteen years ago, he tried to find this code by enumerating the rules and patterns of states of lattice sites of simple cellular automata [73]. In recent years, Wolfram has supplemented these searches by enumerating the rules of transformation graphs of sets of relations of identical but labeled discrete elements [74,75]. The code of the Universe has not yet been found, and Wolfram's project to search for a fundamental theory of physics has been criticized [76]. It is unlikely to find the rules for the creation of the Universe just by combining numbers. Even if cellular automata and graphs are used. We share the idea that our Universe is based on cellular automaton logic, but we believe that a more meaningful search for an appropriate fundamental theory is needed. It is necessary to find the first principles (objects and axioms) of the general physical theory and put these principles into the rules of the cellular automaton of our Universe. Without creating a common ontology of our world, it is impossible to invest in AI algorithms for expedient, safe and useful behavior.

3.3. Cellular-automaton nature of board games on a cellular field

The board games develop logical and imaginative thinking skills in the field of strategy and tactics. Chess is the most famous logic game that simulates the battle of two armies on the field. The most ancient of the currently known strategic games on the cellular field are Senet and the Royal game of Ur. The game of Senet was invented in ancient Egypt about 5500 years ago. The royal game of Ur was invented in Sumer at least 4400 years ago.

We believe that in order to create a board game on a cellular field, it is first necessary to create a theory based on the first principles. To create a theory means to define the basic objects of the theory (pieces, board, starting pattern) and the axioms of the behavior of these basic objects (rules of the game). This classic rational approach to modeling emerged spontaneously and independently in different civilizations. The prevalence of logical board games on the cellular field in all advanced civilizations testifies to their unique effectiveness for modeling complex systems and the exciting interest that they aroused among the players, providing full visibility and transparency of all events on the field. We consider these games to be cellular automata because they have similar features: (1) the field consists of ordered sites (cells), (2) each site can be in a finite number of known states, (3) the pieces have specific local neighborhoods, (3) there is an initial field pattern (4) transitions between site states are implemented iteratively according to certain logical rules. Given the prevalence of cellular automaton stereotypes in board games of different civilizations, it can be assumed that cellular automaton thinking is natural for humans. Perhaps our thinking uses cellular automaton modeling of the world we live in. At the same time, human imagination is not enough to use all the game pieces of the opposing sides at each iteration of the game. As a result, the game is played by alternating single moves of individual pieces. Computers opened up the possibility of modeling changes in the states of all sites of the field at each iteration, which increased the realism of the models, since in nature, many processes occur simultaneously. The board games on a cellular field create a simulation of reality. Their spontaneous cellular automata nature potentially confirms the idea that the Universe is a cellular automaton.

3.4. From the metaphors of the game of life to precedents for the implementation of completely transparent models

The computer cellular automaton game "Life" (Game of Life) was invented by John Conway [77]. The player arbitrarily creates a pattern of "live" and "dead" sites on a two-dimensional field, and then only observes the evolution of the patterns. If the number of "living" neighboring sites is 2 or 3 out of 8, then at the next iteration the site "survives", if not, then it dies. A "dead" site "comes to life" at the next iteration if only it is surrounded by three "live" sites. The player hopes to get interesting dynamics during the evolution of the machine. Of course, this is not a model of a specific object of nature, but a very distant metaphor that can be attributed to abstract art. But this metaphorical model has analogies in many disciplines and allows you to reproduce dynamic processes that were not amenable to mathematical modeling before. This model is important to us because we used its program [78] as a basis, replacing its rules with those of competing species while considering the objects and axioms of general ecosystem theory. The ecological model of resource competition that we have implemented has a completely transparent nature, unlike all available solutions, and has made it possible to solve a

number of problems of theoretical ecology by the XXAI approach [24,40,41]:

1. Novel, fully transparent individual-based mechanisms for the formation of classic population growth curves, including the double-S curve [39].
2. For the first time, a discrete model of population catastrophes has been created; it has been demonstrated that with an increase in the recovery time of ecosystem resources, a catastrophic death of the population occurs [79].
3. Mechanisms have been found for the indefinitely stable coexistence of complete resource competitors under conditions under which coexistence was previously considered impossible. This solved the paradox of biodiversity and opened up new ways to preserve biodiversity [40–42].
4. Two contradictory hypotheses of limiting similarity and limiting dissimilarity of competing species were tested, the hypothesis of limiting similarity was rejected and the hypothesis of limiting dissimilarity was confirmed [24].
5. For the first time, the competitive exclusion principle was rigorously tested and reformulated, and a generalized formulation of the competitive exclusion principle for an arbitrary number of resource competitors was given [40–42].
6. For the first time, the general principle of competitive coexistence was formulated for an arbitrary number of resource competitors [40,42].

3.5. The rules of our cellular automata models in brief

One individual can occupy only one microhabitat. The life cycle of an individual, like the state of regeneration of a microhabitat, lasts one iteration of the cellular automaton. All states of all sites have the same duration. Each individual of all species consumes the same number of identical resources, i.e. they are identical consumers. Such species are complete competitors. Individuals are immobile at the lattice sites and population waves propagate due to the reproduction of individuals. A neighborhood consists of a site and its neighboring sites. All sites have the same update rules. The neighborhood models the meso-habitat of the individual and determines the individual's fertility (the number of possible descendants). More detailed descriptions of our cellular automaton models can be found in our papers, particularly the ones cited in this paper. One particularly relevant paper is [24], which presents the general theory of an ecosystem with competing species and includes basic source code.

3.6. Environmental aspect of symbolic AI decision modeling

The key feature of the proposed approach is that the problem situation solved by AI is modeled as a multi-agent ecosystem, the subsystems of which are isolated micro-ecosystems of individual agents. This isolation is actualized at the moment of making a decision about changing the state by each individual agent at each cycle of the cellular automaton. The meaning of this isolation is to take into account *all* the circumstances that are relevant for the expedient behavior of the agent at the moment of making this decision. The circumstances that are taken into account are part of the agent's cellular automaton neighborhood. For the future, we would like to note that not only each agent of the cellular automaton model can have a specific local neighborhood, but this neighborhood can also change depending on the circumstances that are identified by the sensory capabilities of the agent.

3.7. Cellular automata implement multilevel "top-down" and "bottom-up" approaches to modeling complex systems

Cellular automata are "bottom-up" models that generate global behavior from local rules [80]. Through bottom-up inference, we able to ensure that our model mechanistically constructs the ecosystem at each iteration from the local conditions of each individual microhabitat and each individual of the species to the ecosystem as a whole. The rules of cellular automata are implemented at three levels of ecosystem organization. The possibility of multilevel modeling is one of the transparency aspects of our method. The microlevel is modeled by a lattice site (microecosystem). The mesolevel of local interactions of a microecosystems is modeled by the neighborhood of cellular automata (mesoecosystem). The macro level (the whole ecosystem) is the entire cellular automaton (lattice shape and size, boundary conditions, cellular automaton neighborhood, general cellular automaton rules for state transitions), and the parts are the sites and their neighbors. The logical rules of the model include both "part-whole" relationships and "cause-effect" relationships. This approach combines discreteness of microecosystems and their states with explicit consideration of their cause-effect relationships in modeling ecosystem dynamics.

Creation of a transparent mathematical model assumes knowledge of the algorithms for reproducing the system by logically combining its elements into an integral system. Transparency also implies knowledge of the algorithms that determine the mechanisms of the system's functioning. Understanding the mechanism implies the ability to create in the imagination or on a computer its transparent mathematical model, i.e., the transparent model of the system is its integral mechanism.

The key feature of mathematical model transparency is the requirement for simultaneous implementation of cause-and-effect algorithms of the "part-whole" relationship, both relatively "top-down" and relatively "bottom-up". This requirement applies to both operational and semantic transparency. If operational transparency is achieved by deterministic logical cellular automata, then semantic transparency must be provided by the general theory of the relevant domain. The latter requirement is problematic because of the partial loss of the culture of rational thought in modern positivist science. The simultaneous implementation of top-down and bottom-up approaches in multilevel cellular automata models eliminates the problem of confrontation between reductionist and holistic approaches.

4. Conclusions and prospects

4.1. Explicitly Explainable AI (XXAI)

To overcome all four barriers to the widespread use of symbolic AI and to obtain a white box solution, we propose the use of logically deterministic cellular automata whose rules are based on the general physical theory of the relevant domain. This is the main hypothesis of this paper. The comprehensive analysis of this hypothesis carried out here fully confirmed it and allowed to make predictions about the prospects of implementation of the proposed solution.

This approach opens up great prospects for various theoretical and applied areas. The artificial intelligence developers have gained a precedent for using fully transparent and fully explainable artificial intelligence to model the behavior of multi-particle and multi-level systems. Rational logical artificial intelligence brings with it the explainability, reliability, controllability and security. It is possible for AI not only to process data statistically, as in today's successful AI systems, but also to work directly with knowledge. This work provides the methodological foundation for a research and development program to create general-purpose symbolic AI as the Explicitly Explainable AI (XXAI).

4.2. Explicitly Explainable AI (XXAI) Creation Program

We expect that the central result of this program will be the creation of a unified base of logical scenarios for rational agent behavior, based on first principles of general theories from various domains. Scientists and developers will be able to use these algorithms directly when creating symbolic AI systems. The main thing that will be needed to create such a foundation is a universal ontology of explicit knowledge

[52,55,56]. It is necessary to find a universal ethical foundation (goal function) that is common to all humans and all forms of AI. We propose to use the general theory of living systems as the basis for such a universal ontology [60,61]. On the basis of axioms of general theories of different fields, a unified basis of first principles for fully transparent scenarios of rational behavior of any agent will be created.

4.3. Our recommendations for future work on the development of XXAI

There is a certain difficulty in attracting the implementers of such a program due to the 150-year dominance of positivism in science. Rational knowledge is based on supersensible concepts. Supersensible concepts are insufficiently familiar objects for the work of modern scientists, who are accustomed to obtaining and publishing reliable experimental data. The path from experimental data to rational knowledge is through reflection, which is often perceived by reviewers as unacceptable speculation [64,65]. The ship of modern science needs to make a course correction so that the traditions of classical principles of scientific rationality find their place in the way we know the world around us. Scientists should see themselves not only as miners extracting data, but also as Euclids creating new axiomatic theories of their fields. Of course, it is necessary to understand that axioms cannot be arbitrary, but only the result of a sincere, deep and honest immersion in the respective subject. Only when the assimilation of the facts of a given subject is sufficiently complete, it is possible to achieve its internal understanding through persistent reflection on the available facts and the explanatory concepts of its predecessors. This internal model of the domain is the basis for creating the corresponding basic objects and axioms of general theories.

Successful work with first principles is facilitated by a combination of genuine scientific curiosity, a propensity for theorizing (the ability to produce results "at the tip of the pen" through reflection), an understanding of the foundations of classical scientific rationality, and a deep immersion in the subject matter. A combination of such qualities and knowledge was often possessed by the developers of theoretical biology, inspired by the classic works of E. Schrödinger ("What is Life"), J. von Uexküll ("Theoretical Biology"), C. H. Waddington ("Towards a Theoretical Biology"), I. Prigozhin (theory of dissipative self-organization), E. Bauer ("Theoretical Biology") … Nowadays, such research and introducing students to it is important to maintain and improve the culture of theoretical thinking in science. The importance of theoretical biology research is also related to the fact that it tries to find a global goal function of living systems, which seems to be common to both man and machine.

In addition to general theoretical work to ensure clarity of the concepts used, a program to develop symbolic AI for a wide range of purposes should include comprehensive development of methods for modeling cellular automata. Each simulated agent in the cellular automata paradigm must be able to rationally adapt its behavior to the dynamics of its ecosystem before each iteration of the model. Hierarchical network co-organization of agents must be ensured, and transparent AI algorithms must be coupled with modern geographic information systems. The use of category theory, which has created the richest language for modeling complex systems, seems very promising [81–83]. Category theory is the most general theory of mathematical structures. It applies to any kind of mathematical structure, and also studies the relations between structures of different kinds. Category theory and cellular automata are united by their reliance on abstractions of states of structures and transitions between these states. We believe it is important to introduce the language of category theory to developers of cellular automata models.

XXAI, based on cellular automata is multi-component, multi-level, networked, and hyper-logical in nature. These rich capabilities offer great potential for developing advanced autonomous solutions for a prosperous and sustainable industrial future.

CRediT authorship contribution statement

Vyacheslav L. Kalmykov: Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Methodology, Investigation, Conceptualization. **Lev V. Kalmykov:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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